

CPU & Memory Unit – Page 491

Source: Lucent's General Knowledge – Computer

Computer: CPU Components & Memory Types

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. The CPU is the most important part of the computer. It is called:

- A. Heart of computer
- B. Brain of the computer
- C. Eyes of computer
- D. Hands of computer

Ans: B

Q2. The CPU can be divided into three main components:

- A. RAM, ROM, Cache
- B. ALU, CU and Registers
- C. Input, Output, Storage
- D. Monitor, Keyboard, Mouse

Ans: B

Q3. The Arithmetic and Logic Unit (ALU) performs:

- A. Only storage
- B. All the mathematical and logical operations on the information supplied to the CPU
- C. Only printing
- D. Only networking

Ans: B

Q4. Control Unit (CU) directs the working of the CPU by fetching instructions from:

- A. Output devices
- B. Memory
- C. Printer
- D. Monitor

Ans: B

Q5. Registers are storage locations that hold instructions or data while the CPU is using them. They consist of:

- A. Transistors
- B. Flip-flops and registers used by the CPU
- C. Capacitors
- D. Inductors

Ans: B

Q6. CPU registers are the fastest memory elements in:

- A. Hard disk
- B. The computer
- C. Internet
- D. Printer

Ans: B

Q7. Data Width refers to:

- A. Physical size of CPU
- B. Number of bits of data that can be manipulated within the CPU at one given time
- C. Weight of computer
- D. Length of wires

Ans: B

Q8. The data width of a computer is also called its: A. Byte size B. Word size C. File size D. Disk size	Ans: B
Q9. Computers have data widths ranging from: A. 1 to 4 bits B. 8 to 64 bits C. 100 to 200 bits D. 1000 bits	Ans: B
Q10. A higher data width means the CPU is capable of processing data: A. Slower B. Faster C. Same speed D. Cannot process	Ans: B
Q11. Address Range refers to the amount of memory that can be: A. Printed B. Directly read or written by the CPU C. Deleted D. Copied only	Ans: B
Q12. The speed of CPU is known as: A. Baud Rate B. Clock Speed C. Bandwidth D. Frequency	Ans: B
Q13. The computer is essentially composed of tiny devices that can be put on or off to indicate: A. Colors B. 1 or 0 C. Sound D. Light	Ans: B
Q14. To synchronize the change of all components, the CPU uses: A. External clock B. An internal clock C. Alarm D. Timer	Ans: B

Q15. The memory unit stores all instructions and data for: A. The printer B. The CPU C. The monitor D. The keyboard	Ans: B
Q16. Memory is of two kinds: A. Fast and Slow B. Primary and Secondary C. Big and Small D. Old and New	Ans: B
Q17. Primary Memory is often referred to as the working memory or: A. Storage memory B. Main memory of a computer system C. External memory D. Backup memory	Ans: B
Q18. Primary memory is temporary in nature, i.e. Data stored in primary memory is lost when: A. New data comes B. The computer is switched off C. The disk is full D. Internet disconnects	Ans: B
Q19. Primary memory is also called volatile memory. Example of primary memory is: A. Hard disk B. RAM C. CD D. USB	Ans: B
Q20. The two basic kinds of primary memory are: A. HDD and SSD B. Random Access Memory (RAM) and Read Only Memory (ROM) C. CD and DVD D. USB and SD Card	Ans: B
Q21. RAM is a: A. Read only memory B. Read/write memory C. Non-volatile memory D. Permanent memory	Ans: B

Q22. RAM is volatile, which means: A. Data is permanent B. Data is lost when power is switched off C. Data cannot be written D. Data is always backed up	Ans: B
Q23. RAM capacity greatly influences the: A. Weight of computer B. Computing ability of the computer C. Color of monitor D. Size of keyboard	Ans: B
Q24. The ROM can NOT be: A. Read B. Altered C. Powered D. Installed	Ans: B
Q25. Information is stored on the ROM at the time of its: A. First use B. Manufacture C. Installation D. Update	Ans: B
Q26. The ROM is non-volatile and retains its information even after: A. Data deletion B. The power is turned off C. Formatting D. Virus attack	Ans: B
Q27. PROM stands for: A. Permanent Read Only Memory B. Programmable Read Only Memory C. Primary Read Only Memory D. Protected Read Only Memory	Ans: B
Q28. Cache is the primary memory that stores: A. Permanent data B. Temporary data for quick access C. Only images D. Only videos	Ans: B
Q29. Secondary Memory is used to store data for: A. Very short time B. A long term C. Only during shutdown D. Only during startup	Ans: B

Q30. Secondary memory is permanent in nature, so it is also called: A. Volatile memory B. Non-volatile memory C. Temporary memory D. Flash memory	Ans: B
Q31. Common examples of secondary memory include: A. RAM and ROM B. Floppy diskettes, hard discs, magnetic tapes, CDs, DVDs, Memory Cards, USB Sticks C. Only RAM D. Only Cache	Ans: B
Q32. Secondary memory is NOT directly accessible to the CPU. Information is moved from secondary to: A. Output device B. Primary memory first and then to the CPU C. Printer D. Internet	Ans: B
Q33. A floppy diskette is a plastic disk coated with: A. Glass B. Magnetic material C. Gold D. Silicon	Ans: B
Q34. Special devices known as disk drives are capable of reading from and writing to floppies using: A. Laser B. Special magnetic 'head' C. Electrical current D. Light beam	Ans: B
Q35. Magnetic tapes are long plastic tapes coated with: A. Silicon B. Magnetic material C. Glass D. Copper	Ans: B
Q36. A problem with magnetic tapes is that information can NOT be: A. Stored B. Accessed directly (like in floppy diskettes) C. Written D. Erased	Ans: B

Q37. The third type of medium, called fixed or hard disks, contains: A. Single disk B. Several discs of a hard material C. Paper D. Only one platter	Ans: B
Q38. Primary memory is fast but expensive. To reduce storage costs, computers also use: A. More primary memory B. Secondary memory C. No memory D. Only cache	Ans: B
Q39. Any piece of information stored on a floppy diskette can be: A. Never accessed B. Directly accessed C. Only read once D. Only written once	Ans: B
Q40. The memory unit holds instructions and data: A. During printing B. Before or after the CPU processes these C. During shutdown D. Only at startup	Ans: B
Q41. The Control Unit controls the flow of data between the ALU and: A. Printer B. Other parts of the computer C. Monitor only D. Speaker only	Ans: B
Q42. Primary memory must be able to provide data: A. Slowly B. Very quickly C. Only when requested D. Only once	Ans: B
Q43. Primary memory is directly accessible to: A. The printer B. The CPU C. The monitor D. The keyboard	Ans: B

Q44. The information on ROM might be in the form of crucial instructions that govern the:

- A. Printer
- B. Working of the computer
- C. Internet connection
- D. Speaker volume

Ans: B

Q45. In PROM, the manufacturer of computer may choose to load a program designed by his company. The computer would use this PROM like:

- A. RAM
- B. Any other ROM
- C. Hard disk
- D. Cache

Ans: B

Q46. Magnetic tapes can store far larger amounts of data than:

- A. Hard disks
- B. Floppy diskettes
- C. SSDs
- D. DVDs

Ans: B

Q47. It makes all the required calculations and processes data. This refers to:

- A. Memory Unit
- B. CPU (it makes all the required calculations and processes data)
- C. Output Unit
- D. Input Unit

Ans: B

Q48. The Storage device of a computer system is known as:

- A. CPU
- B. Memory
- C. ALU
- D. CU

Ans: B

Q49. Memory Unit can receive data, hold it and deliver according to the instructions from:

- A. Output unit
- B. Control unit
- C. Printer
- D. Monitor

Ans: B

Q50. With every tick of the internal clock, all switches change their position in:

- A. Random order
- B. Perfect harmony
- C. No particular order
- D. Reverse order

Ans: B

Answer Key & Explanations

Q1. Answer: B — Brain of the computer

Page 491: The Central Processing Unit or CPU is the most important part of the computer. It is called the brain of the computer.

Topic: CPU

Q2. Answer: B — ALU, CU and Registers

Page 491: The CPU can be divided into three main components: (a) ALU, (b) CU and (c) Registers.

Topic: CPU

Q3. Answer: B — All the mathematical and logical operations on the information supplied to the CPU

Page 491: ALU performs all the mathematical and logical operations on the information supplied to the CPU.

Topic: CPU

Q4. Answer: B — Memory

Page 491: Control Unit directs the working of the CPU. It fetches instructions (Programs) from the memory.

Topic: CPU

Q5. Answer: B — Flip-flops and registers used by the CPU

Page 491: Registers consist of flip-flops and registers used by the CPU. They are the fastest memory elements in the computer.

Topic: CPU

Q6. Answer: B — The computer

Page 491: The registers used by the CPU are the fastest memory elements in the computer.

Topic: CPU

Q7. Answer: B — Number of bits of data that can be manipulated within the CPU at one given time

Page 491: Data Width: It refers to the number of bits of data that can be manipulated within the CPU at one given time.

Topic: CPU

Q8. Answer: B — Word size

Page 491: The data width of a computer is also called its word size.

Topic: CPU

Q9. Answer: B — 8 to 64 bits

Page 491: Computers have data widths ranging from 8 to 64 bits.

Topic: CPU

Q10. Answer: B — Faster

Page 491: A higher data width means the CPU is capable of processing data faster. A CPU with higher data width is more powerful.

Topic: CPU

Q11. Answer: B — Directly read or written by the CPU

Page 491: Address range refers to the amount of memory that can be directly read or written by the CPU.

Topic: CPU

Q12. Answer: B — Clock Speed

Page 491: Clock Speed: The speed of CPU is known as Clock Speed.

Topic: CPU

Q13. Answer: B — 1 or 0

Page 491: The computer is essentially composed of tiny devices that can be put on or off to indicate 1 or 0.

Topic: CPU

Q14. Answer: B — An internal clock

Page 491: To synchronize the change of all these components the CPU uses an internal clock.

Topic: CPU

Q15. Answer: B — The CPU

Page 491: The memory unit stores all instructions and data for the CPU.

Topic: Memory

Q16. Answer: B — Primary and Secondary

Page 491: Memory is of two kinds: (a) Primary and (b) Secondary.

Topic: Memory

Q17. Answer: B — Main memory of a computer system

Page 491: Primary Memory is often referred to as the working memory or the main memory of a computer system.

Topic: Memory

Q18. Answer: B — The computer is switched off

Page 491: Primary memory is temporary. Data is lost when computer is switched off. So it is also called volatile memory.

Topic: Memory

Q19. Answer: B — RAM

Page 491: Primary memory is volatile. Example of primary memory is RAM.

Topic: Memory

Q20. Answer: B — Random Access Memory (RAM) and Read Only Memory (ROM)

Page 491: Two basic kinds of primary memory are Random Access Memory (RAM) and Read Only Memory (ROM).

Topic: Memory

Q21. Answer: B — Read/write memory

Page 491: The RAM is a read/write memory. The CPU can change the contents of the RAM at any time.

Topic: Memory

Q22. Answer: B — Data is lost when power is switched off

Page 491: RAM is volatile. In addition, RAM is volatile - data lost when computer switched off.

Topic: Memory

Q23. Answer: B — Computing ability of the computer

Page 491: The RAM capacity greatly influences the computing ability of the computer. Capacity usually measured in kilobytes and megabytes.

Topic: Memory

Q24. Answer: B — Altered

Page 491: The ROM can not be altered. Information stored on ROM at time of manufacture.

Topic: Memory

Q25. Answer: B — Manufacture

Page 491: Information is stored on the ROM at the time of its manufacture.

Topic: Memory

Q26. Answer: B — The power is turned off

Page 491: The ROM is non-volatile and retains its information even after the power is turned off.

Topic: Memory

Q27. Answer: B — Programmable Read Only Memory

Page 491: PROM (Programmable Read Only Memory) has the option of being programmed.

Topic: Memory

Q28. Answer: B — Temporary data for quick access

Page 491: Cache is the primary memory that stores temporary data for quick access.

Topic: Memory

Q29. Answer: B — A long term

Page 491: Secondary Memory: It is used to store data for a long term. It operates at a much slower rate than primary memory.

Topic: Memory

Q30. Answer: B — Non-volatile memory

Page 491: Secondary memory is permanent in nature, so it is also called non-volatile. It is also cheaper than primary memory.

Topic: Memory

Q31. Answer: B — Floppy diskettes, hard discs, magnetic tapes, CDs, DVDs, Memory Cards, USB Sticks

Page 491: Common examples of secondary memory are floppy diskettes, hard (fixed) discs and magnetic tapes, CDs, DVDs, Memory Cards, USB Sticks.

Topic: Memory

Q32. Answer: B — Primary memory first and then to the CPU

Page 491: It is not directly accessible to CPU. Information is moved from secondary memory to primary memory first and then to CPU.

Topic: Memory

Q33. Answer: B — Magnetic material

Page 491: A floppy diskette is a plastic disk coated with magnetic material.

Topic: Memory

Q34. Answer: B — Special magnetic 'head'

Page 491: Special devices known as disk drives are capable of reading from and writing to floppies using special magnetic 'head'.

Topic: Memory

Q35. Answer: B — Magnetic material

Page 491: Magnetic tapes are long plastic tapes coated with magnetic material.

Topic: Memory

Q36. Answer: B — Accessed directly (like in floppy diskettes)

Page 491: A problem with magnetic tapes is that information can not be accessed directly as in floppy diskettes.

Topic: Memory

Q37. Answer: B — Several discs of a hard material

Page 491: The third type of medium, called fixed or hard disks, are more or less similar to floppy. One hard disk drive contains several discs of hard material.

Topic: Memory

Q38. Answer: B — Secondary memory

Page 491: Primary memory is fast but expensive. To reduce storage costs, computers also use secondary memory.

Topic: Memory

Q39. Answer: B — Directly accessed

Page 491: Any piece of information stored on a floppy diskette can be directly accessed.

Topic: Memory

Q40. Answer: B — Before or after the CPU processes these

Page 491: The memory unit holds instructions and data before or after the CPU processes these.

Topic: Memory

Q41. Answer: B — Other parts of the computer

Page 491: Control Unit controls flow of data between the ALU and other parts of the computer.

Topic: CPU

Q42. Answer: B — Very quickly

Page 491: Primary memory is directly accessible to the CPU. It must be able to provide data very quickly.

Topic: Memory

Q43. Answer: B — The CPU

Page 491: Primary memory is directly accessible to the CPU.

Topic: Memory

Q44. Answer: B — Working of the computer

Page 491: Information on ROM might be crucial instructions that govern the working of the computer.

Topic: Memory

Q45. Answer: B — Any other ROM

Page 491: In PROM, manufacturer may load a program. The computer would use this PROM like any other ROM.

Topic: Memory

Q46. Answer: B — Floppy diskettes

Page 491: Magnetic tapes can store far larger amounts of data than the floppy diskette.

Topic: Memory

Q47. Answer: B — CPU (it makes all the required calculations and processes data)

Page 491: CPU makes all the required calculations and processes data.

Topic: CPU

Q48. Answer: B — Memory

Page 491: The storage device of a computer system is known as memory.

Topic: Memory

Q49. Answer: B — Control unit

Page 491: Memory Unit can receive data, hold it and deliver according to the instructions from the control unit.

Topic: Memory

Q50. Answer: B — Perfect harmony

Page 491: With every tick of this clock all switches that need to change their position do so in perfect harmony.

Topic: CPU